To find a reasonable value for k using the "elbow method", we calculated the inertia for a range of cluster counts from 2 to 30. Inertia measures how tightly clustered the data points are; lower inertia indicates that points are closer to their respective centroids.

Observations and Analysis

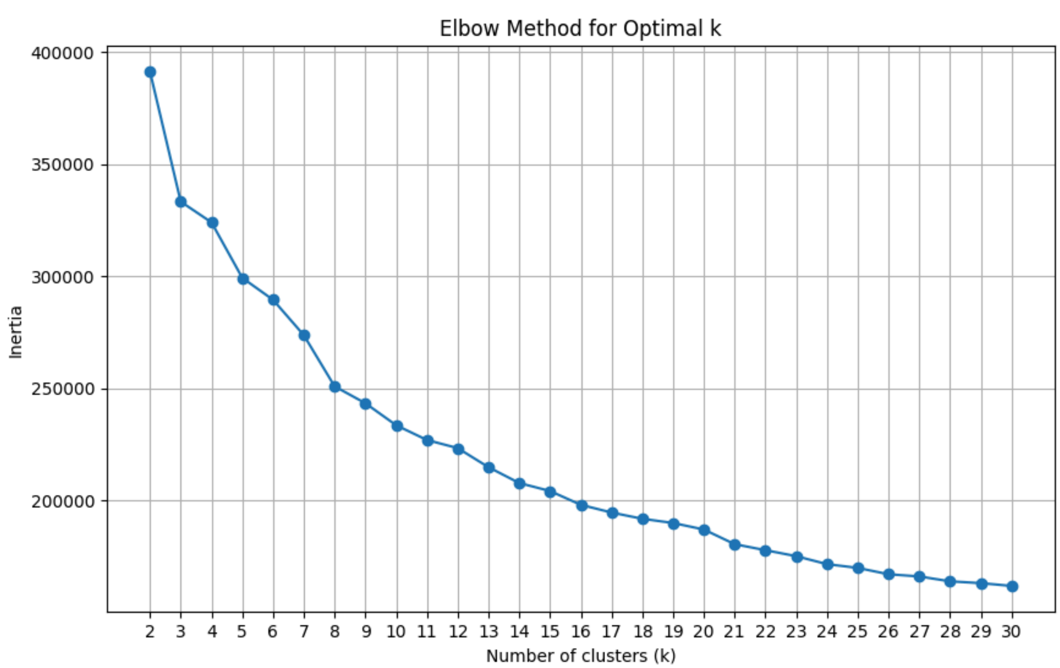
1. **Inertia Trend**: The inertia decreases as k increases, which is expected since more clusters will naturally be closer to their member points.
2. **Identifying the "Elbow"**: The "elbow" is the point on the graph where the rate of decrease in inertia slows down significantly. In this dataset, there isn't one singular sharp elbow, but we can see noticeable changes in the slope:
   - There is a significant drop from k=2 to k=3.
   - Another noticeable shift occurs around k=8 to k=10.
   - After k=15, the curve becomes much flatter, suggesting that adding more clusters beyond this point provides diminishing returns in explaining the variance of the data.

Selecting k

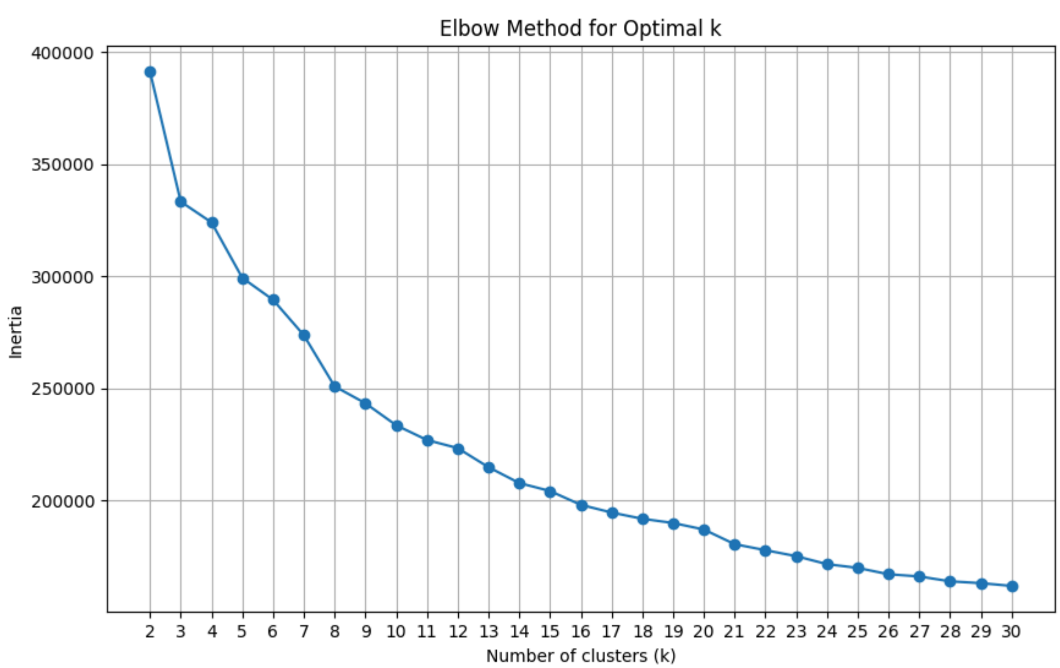
Based on the plot, a value of **k=8** or **k=13** would be a reasonable choice. These points represent areas where the "elbow" begins to form, balancing model complexity (number of clusters) with the tightness of the clusters (inertia).

Summary of Results

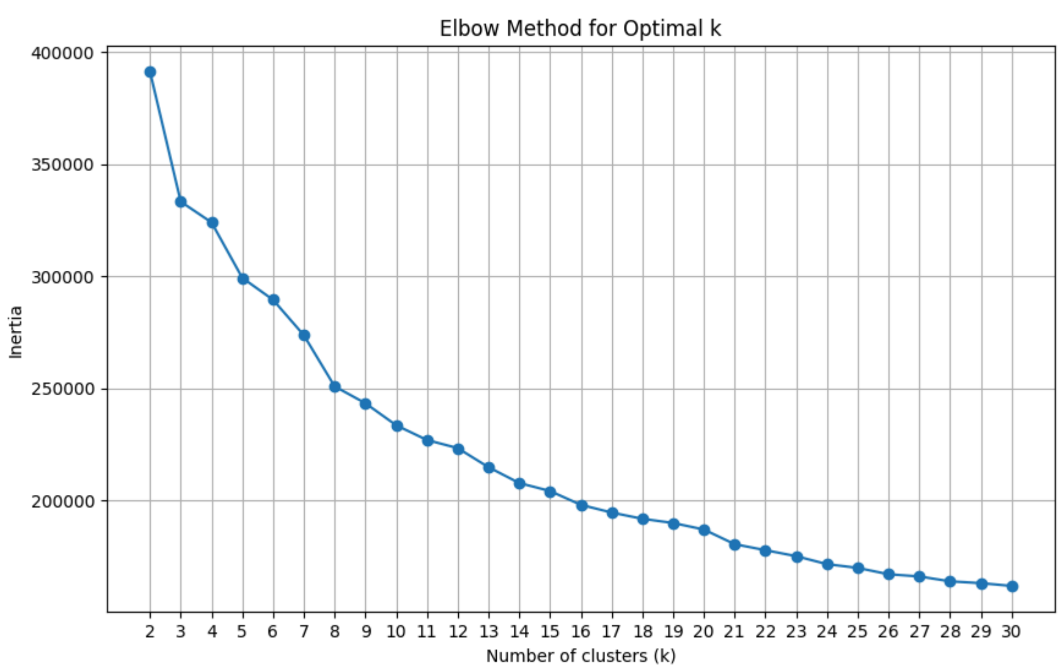
The calculated inertia values for each k are provided in the table below (abbreviated):



k	Inertia
2	391,327.87
5	299,330.01
8	250,986.97
10	233,653.10
15	204,334.65
20	187,175.88



30	162,039.05



Elbow Method for Optimal k
400000
350000
300000
250000
200000
Inertia
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
Number of clusters (k)